

Atho Like Bitcoin

A repository-grounded technical and economic comparison of Bitcoin and Atho

Protocol Architecture, Consensus Rules, and Reference Implementation

Pre-production notice: Atho is experimental software under active development. This paper describes the current repository; it is not a mainnet launch approval, investment statement, or claim of completed independent security assurance.

Version context: Atho v0.1.0 repository state

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Abstract

Bitcoin remains the benchmark proof-of-work digital asset because it combined a scarce monetary policy, a public UTXO ledger, and independently verifiable full-node validation into a system that has survived real adversarial conditions for many years. Atho is clearly inspired by that same family of design choices. It is a proof-of-work, UTXO-based payment network that prefers simple settlement logic over a broad smart-contract surface. At the same time, Atho deliberately diverges from Bitcoin in several major areas: it uses SHA3-384 for proof-of-work rather than SHA-256d, Falcon-512 transaction signatures rather than elliptic-curve signatures, a four-era bootstrap emission schedule with a permanent 0.50 ATHO tail reward instead of a finite hard cap, a different nominal supply path while retaining eight-decimal atom precision, a narrow canonical lock format rather than Bitcoin Script's broader spend policy model, an adaptive 1 MB to 10 MB block-size rule, and a wallet-side transaction proof-of-work anti-spam layer.

This document compares Bitcoin and Atho system by system using the current Atho repository rather than marketing claims. The main finding is that Atho is best understood as a modern, more opinionated Bitcoin-style payment chain rather than as a direct Bitcoin clone. In some areas Atho is simpler than Bitcoin, especially around address handling, ownership locking, and post-quantum signature strategy. In other areas Bitcoin remains much stronger, especially in real-world decentralization, monetary credibility, infrastructure maturity, standardization, and long-term adversarial proof. Atho appears to be attempting to solve a different mix of problems: accessibility, lower-friction payments, stable bootstrap-and-tail miner incentives, explicit multi-network isolation, and a cleaner cryptographic transition away from elliptic-curve signatures.

The comparison also identifies several places where Atho deliberately diverges from Bitcoin policy. In the current repository, Atho now implements a hybrid transaction policy: consensus requires every non-coinbase transaction to pay at least $\max(600 \text{ atoms}, 1 \text{ atom per serialized byte})$, TX-PoW remains the required anti-spam gate, and miners may collect optional positive tips above that floor from included transactions. That makes Atho's economic and anti-spam story materially different from Bitcoin's fee-market-first design.

Keywords: Atho, Bitcoin, proof-of-work, UTXO, Falcon-512, SHA3-384, monetary policy, transaction fees, digital cash

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A Repository-Grounded Technical and Economic Comparison of Bitcoin and Atho

Version context: Atho v0.1.0 repository state

Method: current Atho repository and documentation as source of truth for Atho; Bitcoin white paper, Bitcoin developer reference, and relevant BIPs as primary comparison sources for Bitcoin

Keywords

Bitcoin, Atho, proof of work, UTXO, Falcon-512, SHA3-384, digital cash, monetary policy, transaction anti-spam, post-quantum signatures

Scope And Method

This document uses the attached prompt as a structural guide, but it is intentionally narrower and more conservative than a full 25,000-word academic manuscript. The aim here is to create a repository-native comparison that developers, miners, operators, exchanges, and researchers can use immediately.

For Atho:

- the live repository and its docs are treated as the source of truth
- features are labeled by implementation status
- proposed features are not presented as live facts

For Bitcoin:

- the comparison relies on primary or near-primary technical sources
- the paper focuses on architecture, consensus, transaction policy, and software shape rather than current market prices or temporary network statistics

Atho Status Legend

Every Atho feature in this document is described with one of these statuses:

Status	Meaning
Implemented	Present in the current Atho repository and docs as live behavior
Partially implemented	Some support exists, but the full product claim is not complete
Proposed outside repo	Mentioned in planning or target policy, but not the current repository truth
Not implemented	No live implementation found in the current repo

Executive Summary

At a high level, Atho is like Bitcoin in the places that matter most for base-layer payment architecture:

- proof-of-work ordering
- UTXO-based ownership and spending
- local full-node validation
- public peer-to-peer replication
- coinbase-based issuance
- wallet construction separated from consensus truth

At the same time, Atho departs from Bitcoin in ways that are not cosmetic:

- Atho uses **SHA3-384** instead of Bitcoin's **SHA-256d**
- Atho uses **Falcon-512** transaction authorization instead of **ECDSA/Schnorr over secp256k1**
- Atho uses a **four-era 8/4/2/1 ATHO bootstrap schedule plus a permanent 0.50 ATHO tail reward** instead of Bitcoin's 50 BTC launch subsidy, 210,000-block halvings, and 21 million BTC cap
- Atho uses **8 decimal places**, matching Bitcoin's public precision while using atoms as its smallest unit
- Atho uses **wallet TX-PoW anti-spam friction** instead of relying only on fee pressure
- Atho uses a **narrow canonical lock-digest model** instead of Bitcoin's more expressive script ecosystem

The main tradeoff is straightforward:

- Bitcoin is far more proven, standardized, and economically legible to the world.
- Atho is more modern in implementation style and more opinionated in protocol shape, but much less proven.

1. Bitcoin Overview

Bitcoin was introduced as peer-to-peer electronic cash and later evolved in practice into both a settlement asset and a form of scarce digital reserve money. Its core design includes:

- a public UTXO ledger
- proof-of-work ordering
- a 21 million BTC cap
- a block subsidy that halves roughly every 210,000 blocks
- fee-paying transactions competing for limited block space
- script-based spending rules
- a full-node validation model where each node independently checks blocks and transactions

Modern Bitcoin is no longer only the 2009 design. The current system includes:

- SegWit and weight accounting
- Taproot and Schnorr signatures
- multiple address families including legacy Base58, P2SH, Bech32, and Bech32m
- a robust fee market
- a mature wallet, exchange, custody, explorer, and hardware-signer ecosystem

- second-layer systems such as Lightning

That matters for the comparison. Atho is not only competing with the original Bitcoin paper. It is implicitly being compared with the present reality of Bitcoin as a live, globally integrated monetary network.

2. Atho Overview

Atho presents itself in the repo as a proof-of-work payment network built for node operators, wallet users, miners, and explorers. The repo currently includes a desktop wallet, a standalone node, local RPC, a read-only explorer surface, multi-network support, pruning support, transaction anti-spam proof-of-work, and explicit ruleset/version scaffolding.

The current live Atho profile is:

Area	Current repo state	Status
Consensus family	Proof of Work	Implemented
PoW hash	SHA3-384	Implemented
Transaction signatures	Falcon-512	Implemented
Ledger model	UTXO	Implemented
Decimals	8	Implemented
Smallest unit	1 ATHO = 100,000,000 atoms	Implemented
Genesis emission	0 ATHO at height 0	Implemented
Premine	None	Implemented
Bootstrap issuance	18,750,000 ATHO	Implemented
Maximum supply	None	Implemented
Block reward	8, 4, 2, and 1 ATHO for 1, 250, 000 blocks each, then 0.50 ATHO forever	Implemented
Tail activation	Height 5, 000, 001	Implemented
Coinbase maturity	100 blocks	Implemented
Block cadence	100 seconds	Implemented
Block budget	Adaptive 1,000,000-byte launch limit within 10,000,000-vbyte / 10,000,000-raw-byte hard ceilings	Implemented
TX anti-spam	Required TX-PoW	Implemented
Read-only explorer/API	Local HTTP API plus explorer routes	Implemented
Message sign/verify	Desktop client local signing and verification	Implemented
Coin control and freeze tools	Present in the alpha desktop wallet flow	Implemented
Required transaction fee floor	max(600 atoms, 1 atom per serialized byte) with mandatory TX-PoW	Implemented
Optional miner tips as separate mechanism	Positive input-output remainder can be claimed by coinbase	Implemented

Area	Current repo state	Status
AISL / instant settlement layer	No live implementation found	Not implemented

In spirit, Atho is not trying to be a general-purpose smart-contract system. It is trying to be a settlement and payment chain with a more modern cryptographic stack and a different long-run miner economics story.

3. Current Repo Facts That Match The Live Policy

Topic	Target prompt direction	Current repo truth	Result
Monetary policy	four bootstrap eras plus permanent tail	repo now uses 8/4/2/1 ATHO over four 1,250,000-block eras, then 0.50 ATHO forever	aligned
Genesis emission	zero-emission anchor	genesis block is a zero-subsidy anchor and ordinary miner subsidy starts at height 1	aligned
Supply limit semantics	remove stale cap language	repo now reports bootstrap issuance and tail economics without live max_supply_* fields	aligned
Fees	"minimum fee floor with optional higher tips"	consensus requires max(600 atoms, 1 atom per serialized byte)	aligned
Optional tips	users may optionally tip miners	positive transaction remainder is optional tip income for miners	aligned
Block size uncertainty	compare launch limit versus hard ceiling	current repo launches at a 1,000,000-byte adaptive limit and can grow toward the 10,000,000-vbyte / 10,000,000-raw-byte hard ceilings	finalized in repo
Block time	final 100-second cadence	current repo is explicit at 100 seconds	finalized in repo

Today, Atho is a fee-floor chain with:

- a zero-emission genesis anchor and ordinary subsidy starting at height 1
- a fixed 18,750,000 ATHO bootstrap issuance milestone at tail activation

- a permanent 0.50 ATHO tail reward instead of a terminal hard cap
- a required transaction fee of $\max(600 \text{ atoms}, 1 \text{ atom per serialized byte})$
- miner-collected optional tips above that floor when senders add them
- wallet-side TX-PoW anti-spam friction

4. Monetary Policy Comparison

This is the clearest philosophical split between Bitcoin and Atho.

Bitcoin

Bitcoin's monetary model is built around scarcity, predictability, and eventual subsidy decline:

- 21 million BTC cap
- halvings every 210,000 blocks
- declining subsidy over time
- increasing reliance on transaction fees for miner revenue

This gives Bitcoin an extremely strong scarcity narrative. It also creates a long-run open question: whether fee revenue alone will be enough to sustain the desired level of hashpower and security once subsidy becomes small.

Atho

Atho's current monetary model is a different bootstrap-plus-tail design:

- zero-emission genesis anchor at height 0
- no premine
- ordinary subsidy begins at height 1
- four bootstrap eras of 1, 250, 000 blocks each
- 8, 4, 2, and 1 ATHO bootstrap rewards
- 18, 750, 000 ATHO of bootstrap issuance at height 5, 000, 000
- permanent 0.50 ATHO tail reward from height 5, 000, 001
- no maximum supply limit

This changes the network's economic character in several ways.

What Atho improves or attempts to improve:

- it avoids a single future jump into a fee-only mining regime
- it keeps miner base issuance alive at 157, 680 ATHO per nominal year during the tail phase before fees
- it removes the need for a terminal-dust cap-closing rule
- it preserves a clear bootstrap milestone without pretending that milestone is a hard cap

What Atho gives up versus Bitcoin:

- no Bitcoin-style fixed-scarcity brand anchored to a finite terminal supply

- less historical credibility around the revised monetary schedule
- more dependence on whether users and miners accept perpetual tail emission as a valid long-run tradeoff

The comparison is therefore not "better versus worse" in a simple sense. It is a tradeoff:

- Bitcoin maximizes scarcity signaling.
- Atho prioritizes payment continuity and a permanent miner-security floor over hard-cap branding.

If Atho succeeds, it will likely do so as a chain that treats the bootstrap milestone and the tail reward as equally important parts of one monetary design, not as a failed attempt to imitate Bitcoin's hard cap.

5. Mining And Consensus Comparison

Bitcoin and Atho are both proof-of-work systems, but the mining and validation surfaces are not identical.

Bitcoin

Bitcoin uses:

- SHA-256d proof-of-work
- approximately 10-minute block intervals
- longest-valid-most-work chain selection
- subsidy plus optional miner tips
- a massive industrial mining ecosystem with pools, ASIC specialization, and highly optimized block propagation practice

Bitcoin's consensus is simple in principle but battle-tested in practice. The strongest thing Bitcoin has here is not only design elegance. It is proof under years of real attack, congestion, hardware specialization, governance disputes, and adversarial economic pressure.

Atho

Atho uses:

- SHA3-384 proof-of-work
- 100-second target blocks
- 17-block averaging window and 11-block median-time window for retargeting
- bootstrap-plus-tail base subsidy plus transaction fees
- TX-PoW for transaction-level anti-spam before a transaction even reaches miners

That means Atho is doing two different kinds of work:

- 1 chain-ordering PoW for blocks
- 2 lightweight sender-side PoW for transaction admission friction

This is one of Atho's biggest departures from Bitcoin.

Potential strengths of Atho's design:

- lower-frequency spam can be made more expensive even before blocks are full

- the network does not rely only on fee bidding as its first anti-spam mechanism
- shorter target blocks can improve settlement cadence if propagation remains healthy

Risks and tradeoffs:

- TX-PoW adds wallet complexity that Bitcoin does not require
- low-power devices still have to solve admission work
- the operational behavior of TX-PoW under real congestion remains less proven than Bitcoin's fee market

Bitcoin's model is simpler for senders. Atho's model is more interventionist in the name of anti-spam and accessibility.

6. Transaction Model Comparison

Both systems are UTXO-based, which means both inherit many of the same benefits:

- explicit coin-level accounting
- deterministic input/output validation
- natural support for coin control
- simple replay of ownership history through UTXOs

But the transaction surface is still meaningfully different.

Bitcoin

Bitcoin transactions operate inside a broader script ecosystem. Even if most ordinary users only touch simple address types, the system can encode:

- P2PKH
- P2SH
- SegWit outputs
- Taproot outputs
- multisig and policy scripts through script or descriptors

This flexibility is one reason Bitcoin became such a large ecosystem. It also means Bitcoin's address and output model is historically layered and more complex.

Atho

Atho intentionally narrows the base-layer model:

- spendable outputs use a canonical 32-byte payment digest lock
- the visible address is just an interface encoding for that digest
- non-canonical lock formats are rejected
- the Falcon public key appears later in the witness when the output is spent

This is cleaner than Bitcoin's stacked historical script families, but it is also less expressive.

Bitcoin advantage: far more flexible spend policy ecosystem.

Atho advantage: simpler canonical ownership model with less parser ambiguity.

7. Fees, Anti-Spam, And Mempool Policy

This is the section where marketing language can drift away from implementation, so it is worth being direct.

Bitcoin

Bitcoin's primary anti-spam mechanism is economic:

- limited block space
- mempool admission rules
- fee competition
- miner selection of higher-value transactions when space is scarce

This creates a strong market-based policy surface but can make low-value payments expensive during congestion.

Atho

Atho currently combines:

- a deterministic minimum transaction fee of `max(600 atoms, 1 atom per serialized byte)`
- optional miner tips for congestion priority
- mempool policy and block-space-aware tip estimation
- TX-PoW sender friction

That means Atho is no longer feeless, but it still is not a pure fee-market chain in the Bitcoin sense. Every transaction pays a deterministic minimum fee, TX-PoW remains the anti-spam admission gate, and optional tips remain available as an inclusion-priority tool during congestion.

8. Security Model Comparison

Bitcoin's security strengths

Bitcoin's biggest security advantage is not any single primitive. It is the combination of:

- globally distributed hashpower
- conservative full-node validation
- highly reviewed software
- years of adversarial exposure
- broad infrastructure diversity

Bitcoin also benefits from an enormous amount of operator knowledge. Exchanges, wallets, block explorers, miners, and custody systems already know how to handle Bitcoin correctly.

Atho's security strengths

Atho's strongest security ideas in the repo are:

- narrow consensus objects
- strict canonical validation
- explicit network separation across mainnet, testnet, regnet, and prunetest
- fixed founder-hash metadata in canonical headers
- deterministic versioning and activation scaffolding
- local signing boundaries
- post-quantum-aware Falcon-512 transaction authorization
- replay-oriented storage and pruning recovery tests

In pure design terms, Atho is trying to reduce ambiguity. That is good engineering.

Bitcoin still wins overall

Bitcoin still wins the practical security comparison because:

- its economics are more proven
- its node behavior is more field-tested
- its address ecosystem is more standardized
- its cryptographic and operational stack has been reviewed more intensely and for far longer

So the honest phrasing is:

- Atho may improve some parts of design clarity and cryptographic direction.
- Bitcoin remains stronger in demonstrated, real-world security credibility.

9. Network Architecture Comparison

Bitcoin's P2P network is mature, deeply studied, and broadly deployed. It has evolved through years of behavior around peer discovery, block relay, mempool relay, address relay, and network hardening.

Atho's P2P network in the current repo includes:

- explicit peer management
- handshakes
- address sharing and relay
- bootstrap peers and DNS seeds on testnet
- operator-configured advertise addresses
- real-socket and cluster-style regression coverage in the test suite

That means Atho is already following the same decentralization logic Bitcoin depends on: bootstrap peers help a node join, but peer gossip and direct node-to-node connections are supposed to take over from there.

This is a good sign. It does **not** mean Atho's network is as decentralized as Bitcoin's. It means Atho is built with the right architectural intention.

10. Wallet, Node, Miner, API, And Explorer Comparison

Bitcoin's software ecosystem is much larger than its base node alone. It includes:

- Bitcoin Core
- hardware wallets
- watch-only systems
- custodial and non-custodial wallets
- mining pools
- block explorers
- PSBT and descriptor-based tooling
- Lightning wallets and routing stacks

Atho's current repo is more vertically integrated:

- desktop wallet
- standalone node
- miner binaries
- local RPC
- local HTTP API
- read-only explorer routes
- message signing and verification
- wallet-managed change
- manual coin control and freeze controls in the alpha client

This makes Atho more approachable as a single-repo system. It also means more responsibility is concentrated in one codebase and one release line.

One important production caveat remains:

- the current Atho HTTP API has no built-in auth layer and should not be treated as a public privileged API surface

That does not make the node unusable. It just means Bitcoin's surrounding operational hardening culture is still ahead.

11. Addressing, Privacy Basics, And User Experience

Bitcoin's address world is more mature but more fragmented:

- legacy Base58 addresses
- P2SH addresses
- Bech32 and Bech32m witness addresses
- broad QR, URI, and exchange compatibility

Atho's address model is simpler:

- Base56 visible address
- one leading visible network prefix
- fixed checksum suffix
- canonical 32-byte payment digest under the hood

That is cleaner for a new chain, but Bitcoin still wins on external compatibility.

On basic wallet hygiene, Atho is actually quite modern:

- HD receive and change separation
- fresh address rotation
- custom change validation
- local coin labels
- freeze flags
- manual coin control
- mining reward rotation support

Those are good user-facing tools for a transparent chain. They do not make the chain private in the anonymity sense, but they do make it easier for users to avoid careless clustering.

12. Quantum Resistance And Cryptography Comparison

This is one of Atho's most distinctive claims, and it is a real technical divergence rather than a slogan.

Bitcoin

Bitcoin's modern signature surface relies on elliptic-curve cryptography:

- ECDSA historically
- Schnorr for Taproot-era outputs

Those systems are mature and well understood. Their long-term theoretical weakness is that sufficiently capable quantum systems would threaten elliptic-curve assumptions.

Atho

Atho moves transaction authorization to Falcon-512 and uses SHA3-384 for proof-of-work and hashing commitments. That means Atho is explicitly positioning itself as a post-quantum-aware payment chain.

This is a meaningful architectural difference, but it comes with its own tradeoffs:

- much larger public keys and signatures than Bitcoin
- a younger operational ecosystem
- less historical production proof than secp256k1-based systems

So again the balanced phrasing matters:

- Atho is more ambitious in post-quantum direction.

- Bitcoin is more proven in deployed cryptographic operations.

13. Governance, Upgrade Risk, And Decentralization

Bitcoin does not have formal on-chain governance, but it does have one of the most studied rough-consensus upgrade cultures in software history. Changes are slow, politically expensive, and socially constrained.

Atho's repo currently shows a more explicit code-level upgrade path:

- protocol version
- storage schema version
- ruleset version
- a dormant V2 placeholder with no activation height

The current mechanism is a validated fixed-height schedule, not miner-signaled version bits. Active entries must start at height zero and increase strictly; duplicate names, duplicate versions, reordered heights, and active entries after a dormant placeholder are rejected. This makes future activation points visible and deterministic across reorganizations, but it is not the same thing as Bitcoin's BIP 9 state machine or its mature governance culture.

Bitcoin remains stronger in social decentralization and upgrade legitimacy. Atho is stronger in making dormant upgrade hooks explicit inside the codebase.

Bitcoin hardening principles and Atho equivalents

Bitcoin design	Atho equivalent	Important difference
BIP 30 duplicate transaction protection	Duplicate txids are rejected in each block, live UTXOs cannot be replaced, and historical block recovery is block-scoped.	Atho also requires an exact coinbase height commitment from genesis onward.
BIP 34 coinbase height	The coinbase transaction's <code>lock_time</code> is the exact block height.	Atho uses a fixed canonical u32 field instead of Bitcoin script-number encoding.
BIP 66 strict signature encoding	Falcon keys and signatures require exact lengths and canonical backend encodings.	Falcon-512 replaces DER-encoded ECDSA.
BIP 147 malleability closure	Canonical grouped witnesses, complete input coverage, and a header-committed witness identity close Atho's corresponding ambiguity classes.	Atho has no CHECKMULTISIG dummy stack element.
BIP 9 deployment state	Atho validates a compiled fixed-height ruleset schedule.	No miner-signaled DEFINED/STARTED/LOCKED_IN/ACTIVE/FALSE deployment is active.
BIP 130 header announcements	Bounded direct header batches are contextually validated before block download.	Atho checks SHA3-384 PoW, founder anchors, and its per-block difficulty rule.
BIP 144 witness services	Sync peers must advertise full-block and witness services; stripped data cannot match the witness root.	Atho commits Falcon witness and TX-PoW identity directly in the header.
BIP 152 compact relay	Header-keyed short IDs, collision-safe recovery, strict missing-index responses, and full-block fallback.	Atho short IDs use SHA3-derived keys and full witness identities.

14. What Atho Improves, What Bitcoin Still Does Better

Areas where Atho may improve on Bitcoin's baseline design

- simpler canonical lock model
- modern Rust-first implementation
- explicit repo-level version and activation scaffolding
- post-quantum-aware transaction signatures
- sender-side TX-PoW anti-spam layer

- stronger built-in wrong-network separation across multiple environments
- integrated wallet hygiene features in the alpha client

Areas where Bitcoin still does better

- real decentralization depth
- monetary credibility through the 21 million cap
- exchange and custody integration maturity
- global wallet and hardware support
- battle-tested fee market behavior
- broader spend-policy flexibility
- enormous review surface and adversarial history

15. Final Assessment

The fairest way to describe Atho is this:

Atho is a modern, opinionated Bitcoin-style payment chain that is trying to solve a different mix of long-term problems.

It is like Bitcoin in the essential architecture:

- proof of work
- UTXO validation
- node-verified money
- peer-to-peer replication

It is different from Bitcoin in the core policy decisions:

- a different bootstrap-plus-tail monetary model instead of Bitcoin's 21 million BTC schedule
- Falcon-512 instead of ECC transaction authorization
- TX-PoW sender friction plus optional miner tips
- narrow canonical lock digests instead of Bitcoin's broader script ecosystem

That means Atho should not market itself as "Bitcoin, but better" in a simple universal sense. A more honest and stronger framing is:

- Bitcoin remains the benchmark for proven proof-of-work digital money.
- Atho attempts to be a next-generation payment-oriented digital asset that learns from Bitcoin while changing several foundational assumptions.

If Atho succeeds, it will probably succeed because a community decides it values:

- payment accessibility
- a different bootstrap-plus-tail issuance cadence
- explicit post-quantum direction

- simpler ownership semantics
- a more integrated user-facing stack

more than it values:

- Bitcoin's exact halving-era scarcity brand
- maximal historical proof
- immediate ecosystem compatibility with Bitcoin's world

That is a real thesis. It is also a real tradeoff.

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Appendix A. Atho Feature Status Matrix

Feature	Status	Notes
UTXO ledger	Implemented	Canonical payment-digest locking
SHA3-384 block PoW	Implemented	Live consensus hash family
Falcon-512 transaction signatures	Implemented	Live authorization path
Base56 addresses	Implemented	Network-prefixed visible addresses
8 decimal places	Implemented	1 ATHO = 100,000,000 atoms
Four-era 8/4/2/1 ATHO bootstrap schedule	Implemented	1, 250, 000 blocks per era and 18, 750, 000 ATHO issued before tail activation
Permanent 0.50 ATHO tail reward with no max supply	Implemented	Tail starts at height 5, 000, 001 and continues indefinitely
No premine	Implemented	Live policy
TX-PoW anti-spam	Implemented	Wallet-side admission work
Distinct optional miner tip system	Implemented	Positive transaction remainder can be claimed by coinbase
Minimum fee floor	Implemented	Every non-coinbase transaction pays at least max(600 atoms, 1 atom per serialized byte) before any extra tip.
Desktop coin control and freeze tools	Implemented	Alpha client feature set
Local message signing and verification	Implemented	Desktop settings flow
Read-only explorer and local HTTP API	Implemented	Public privileged exposure still discouraged
Built-in authenticated admin API	Not implemented	Explicit gap in production API guide
AISL / instant settlement layer	Not implemented	No live code found